

Unit 3 Reflection

What?

This week, I explored the foundations of **linear regression** and **linear classification** from a machine learning perspective. One of the most interesting concepts was learning that multivariate linear regression can be interpreted as a simple neural network, as both approaches aim to estimate the optimal parameters of a linear function by minimizing a loss function. In linear regression, the **Mean Squared Error (MSE)** serves as the loss function, and under the assumption of normally distributed errors, minimizing MSE is equivalent to maximizing the likelihood function.

I also examined **linear classification**, focusing on discriminant functions and **logistic regression**. Unlike linear regression, logistic regression is designed to model class probabilities and assumes that the target variable follows a Bernoulli distribution. As a result, **cross-entropy loss** is used instead of MSE, making it more appropriate for classification problems. In addition, I studied several key concepts related to model evaluation, including the trade-off between **Type I and Type II errors**, the impact of classification thresholds on model predictions, and the use of **ROC curves** and **AUC-ROC** as threshold-independent measures of classifier performance.

Through these topics, I gained a deeper understanding of both the theoretical foundations of machine learning models and the importance of selecting appropriate evaluation metrics when assessing their effectiveness.

So What?

This unit highlighted the strong connection between **statistics and machine learning**. Understanding that loss functions are rooted in probability distributions helped me appreciate why different learning tasks require different optimization criteria. In particular, recognizing that MSE is derived from Gaussian assumptions, whereas cross-entropy is based on Bernoulli assumptions, clarified why logistic regression is generally more suitable than linear regression for classification problems.

The discussion of Type I and Type II errors also reinforced the idea that model evaluation is not simply about maximizing accuracy. In many real-world applications, different types of errors have different consequences, meaning that the optimal model often depends on the context in which it will be used. For example, in healthcare screening, failing to identify a positive case may be significantly more costly than generating a false alarm. This highlighted the importance of understanding the practical implications of model decisions rather than relying on a single performance metric.

I found the concept of the **ROC curve** particularly insightful because it illustrates model performance across all possible classification thresholds and clearly demonstrates the trade-off between sensitivity and specificity. Furthermore, learning that **AUC-ROC** provides a threshold-independent measure of a

model's ability to distinguish between classes gave me a more comprehensive perspective on evaluation. This understanding reinforced the importance of selecting metrics that align with the objectives, risks, and costs associated with a specific application rather than relying solely on accuracy as an indicator of success.

Now What?

Going forward, I intend to apply these principles more deliberately when developing predictive models. Rather than treating machine learning algorithms as black boxes, I will pay closer attention to the assumptions underlying their probability models and loss functions. For regression problems, I will evaluate whether the assumptions supporting MSE are appropriate for the data and the problem context. For classification tasks, I will consider logistic regression and cross-entropy loss when probabilistic predictions and class membership estimates are required.

I also plan to move beyond accuracy as my primary performance metric and adopt a more comprehensive evaluation approach. Specifically, I will use **ROC curves** and **AUC-ROC** to assess classification models across a range of thresholds and gain a more complete understanding of their predictive capabilities. In addition, I will consider the practical consequences of **Type I and Type II errors** when selecting classification thresholds, recognizing that the appropriate balance between false positives and false negatives depends on the objectives and constraints of the application.

Ultimately, this unit has strengthened my understanding of the theoretical relationship between statistics and machine learning while providing practical tools for model evaluation and decision-making. Applying these insights will enable me to develop more robust, interpretable, and context-appropriate predictive models in future projects.